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Problem Set 2

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Problem 1

(a) **Theorem**: "Every natural number is tall or wide, but not both."

Answer.

We must show that for every natural number n, n is tall or wide, but not both. Let n be an arbitrary natural number. We proceed by cases.

Case 1: n is tall.
We must show n is not wide. [TODO: using the definition of tall, prove n is not wide.]

Case 2: n is not tall.
We must show n is wide and not both. [TODO: using the definition of tall, prove n is wide.] Since n is wide and not tall, n is not both tall and wide.

In both cases, n is tall or wide, but not both. Since n was arbitrary, every natural number is tall or wide, but not both. ■

(b) **Theorem**: "There exists a happy natural number between 10 and 1000 that has no happy proper divisors."

Answer.

We prove this existentially by choosing $n = 25$ as the witness.

We must verify three things:

(i) 25 is between 10 and 1000: This is clear since $10 < 25 < 1000$. ✓

(ii) 25 is happy. [TODO: using the definition of happy, verify that 25 is happy.]

(iii) 25 has no happy proper divisors. The proper divisors of 25 are 1 and 5. [TODO: using the definition of happy, verify that neither 1 nor 5 is happy.]

Therefore, $n = 25$ is a happy natural number between 10 and 1000 with no happy proper divisors, which proves the theorem. ■

(c) **Theorem**: "There are no devious natural numbers."

Answer.

We prove the equivalent statement: for every natural number n, n is not devious.

Predicate: Let $P(n)$ be the statement "n is not devious."

We proceed by induction on n, with base cases $n = 0$ and $n = 1$.

Base Case ($n = 0$): We must show $P(0)$, i.e., that 0 is not devious. [TODO: using the definition of devious, prove 0 is not devious.]

Base Case ($n = 1$): We must show $P(1)$, i.e., that 1 is not devious. [TODO: using the definition of devious, prove 1 is not devious.]

Inductive Step: Let $n \geq 1$ be an arbitrary natural number. Assume $P(n)$ holds, i.e., that n is not devious. We must show $P(n+1)$, i.e., that $n+1$ is not devious.

Assumption: n is not devious.

Goal: Show that $n+1$ is not devious.

[TODO: using the definition of devious and the inductive hypothesis that n is not devious, prove $n+1$ is not devious.]

By induction, $P(n)$ holds for all $n \in \mathbb{N}$. Therefore, there are no devious natural numbers. ■

Problem 2

(a) Brynmor's mistake is in the inductive step. When substituting the inductive hypothesis into $x_{i+1} = 6x_i - 8x_{i-1}$, he plugs in lower bounds for both x_i and x_{i-1}. However, since x_{i-1} appears with a **negative** coefficient (-8), replacing x_{i-1} with its lower bound 2^{i-1} makes the expression *smaller*, not larger. That is, from $x_{i-1} \geq 2^{i-1}$ one can only conclude $-8x_{i-1} \leq -8 \cdot 2^{i-1}$, so the inequality goes the wrong way:

$$x_{i+1} = 6x_i - 8x_{i-1} \geq 6 \cdot 2^i - 8x_{i-1}$$

is valid, but substituting the lower bound of x_{i-1} on the right gives an upper bound, not a lower bound. Hence the step

$$x_{i+1} \geq 6 \cdot 2^i - 8 \cdot 2^{i-1}$$

is **not justified**.

(b) Erik's mistake is in the base case and the circular reasoning at the end. His inductive hypothesis is $R(i) :=$ "$x_i \geq 3x_{i-1}$," defined only for $i \geq 2$. At the end, he concludes that since $x_1 > 0$ and $x_i \geq 3x_{i-1}$ for all $i \geq 2$, we get $x_i > 0$ for all i. But this final step implicitly assumes $x_{i-1} > 0$ in order to conclude $x_i \geq 3x_{i-1} > 0$, which is exactly what he is trying to prove. That is, the conclusion $x_i > 0$ requires knowing $x_{i-1} > 0$, making the argument **circular**. He never establishes by a separate argument that positivity propagates; he simply assumes it.

(c) Zach's mistake is in the inductive step when $i = 1$. He claims to prove $S(i+1)$ using $S(i)$ and $S(i-1)$, but when $i = 1$, this requires $S(0)$, i.e., $x_0 = 4^0 = 1$. However, the sequence is only defined starting from x_1; x_0 does not exist. So the inductive step **fails for** $i = 1$ because it invokes an undefined term. Indeed, the claim $S(i) :=$ "$x_i = 4^i$" is false for $i = 2$ since $x_2 = 13 \neq 16 = 4^2$, confirming the proof is broken.

(d) We prove that $x_i > 0$ for all $i \geq 1$.

Following Erik's approach, we use the stronger inductive hypothesis

$$P(i) := \text{"}x_i \geq 3x_{i-1} \text{ and } x_{i-1} > 0\text{"}$$

for $i \geq 2$, and handle $x_1 > 0$ separately. Equivalently, we prove by strong induction that $x_i > 0$ for all $i \geq 1$ and $x_i \geq 3x_{i-1}$ for all $i \geq 2$.

More precisely, let $P(i) :=$ "$x_i \geq 3x_{i-1}$" for $i \geq 2$, and we will carry the positivity of all previous terms alongside.

Claim: For all $i \geq 1$, $x_i > 0$. Moreover, for all $i \geq 2$, $x_i \geq 3x_{i-1}$.

Proof by strong induction.

Base Cases:

- $x_1 = 4 > 0$. ✓
- $x_2 = 13 > 0$, and $x_2 = 13 \geq 12 = 3 \cdot 4 = 3x_1$. ✓

Inductive Step: Let $i \geq 2$. Assume that for all $1 \leq k \leq i$, $x_k > 0$, and for all $2 \leq k \leq i$, $x_k \geq 3x_{k-1}$. We wish to show that $x_{i+1} > 0$ and $x_{i+1} \geq 3x_i$.

Consider:

$$x_{i+1} = 6x_i - 8x_{i-1}.$$

By the inductive hypothesis, $x_i \geq 3x_{i-1}$, so $x_{i-1} \leq \frac{x_i}{3}$, and thus $8x_{i-1} \leq \frac{8}{3}x_i$. Therefore:

$$x_{i+1} = 6x_i - 8x_{i-1} \geq 6x_i - \frac{8}{3}x_i = \frac{10}{3}x_i.$$

Since $x_i > 0$ by the inductive hypothesis, we have:

$$x_{i+1} \geq \frac{10}{3}x_i > 3x_i > 0.$$

This establishes both $x_{i+1} \geq 3x_i$ and $x_{i+1} > 0$, completing the inductive step.

By strong induction, $x_i > 0$ for all $i \geq 1$. ■

Problem 3

(a) We compute the sequence of transitions from starting state $(5, 22, 0)$:

r	s	a	Rule applied
5	22	0	s even: $(2r,\ s/2,\ a)$
10	11	0	s odd: $(2r,\ (s-1)/2,\ a+r)$
20	5	10	s odd: $(2r,\ (s-1)/2,\ a+r)$
40	2	30	s even: $(2r,\ s/2,\ a)$
80	1	30	s odd: $(2r,\ (s-1)/2,\ a+r)$
160	0	110	$s = 0$, no transition (final state)

The final state is $(160, 0, 110)$, and indeed $a_f = 110 = 5 \cdot 22$. ✓

(b) We prove that $P(r, s, a) :=$ "$rs + a = xy$" is an invariant of the state machine.

Base case: The start state is $(x, y, 0)$. We verify:

$$r \cdot s + a = x \cdot y + 0 = xy. ✓$$

Inductive step: Suppose (r, s, a) is a reachable state satisfying P, i.e., $rs + a = xy$, and $s > 0$ so a transition is possible. We consider two cases.

Case 1: s is even. The next state is $(2r,\ s/2,\ a)$. We verify P:

$$(2r) \cdot \frac{s}{2} + a = rs + a = xy. \checkmark$$

Case 2: s is odd. The next state is $(2r,\ (s-1)/2,\ a+r)$. We verify P:

$$(2r) \cdot \frac{s-1}{2} + (a+r) = r(s-1) + a + r = rs - r + a + r = rs + a = xy. \checkmark$$

In both cases, P holds for the next state. Therefore P is an invariant. ■

(c) A final state is one from which no transition is defined. By the problem statement, transitions are defined precisely when $s > 0$. Therefore, the final states are all triples of the form

$$(r, 0, a) \quad \text{for any } r, a \in \mathbb{N}.$$

(d) Let (r_f, s_f, a_f) be any reachable final state. By part (c), $s_f = 0$. By part (b), since P is an invariant and the start state satisfies P, every reachable state satisfies P. In particular:

$$r_f \cdot s_f + a_f = xy.$$

Substituting $s_f = 0$:

$$r_f \cdot 0 + a_f = xy \implies a_f = xy.$$

Therefore, in any reachable final state, $a_f = xy$. ■

(e) We prove that bitcount(s) is a strictly decreasing derived variable, i.e., that every transition strictly decreases bitcount(s).

Consider any transition from state (r, s, a) with $s > 0$.

Case 1: s is even. The new value of s is $s/2$. Since $s > 0$, we have $s \geq 2$, so $s/2 \geq 1$. Now:

$$\text{bitcount}(s/2) = \left\lceil \log_2\left(\frac{s}{2} + 1\right) \right\rceil < \lceil \log_2(s+1) \rceil = \text{bitcount}(s),$$

where the inequality holds because $\frac{s}{2} + 1 \leq s + 1$ for $s \geq 2$, so $\log_2(\frac{s}{2} + 1) < \log_2(s+1)$. More directly: dividing s by 2 removes exactly the least significant bit of its binary representation, strictly reducing the number of bits by at least 1.

Case 2: s is odd. The new value of s is $(s-1)/2$. Since $s \geq 1$ is odd, we have $s - 1 \geq 0$ is even, so $(s-1)/2 \in \mathbb{N}$. We verify that the bit count strictly decreases:

$$\begin{aligned}
\text{bitcount}\left(\frac{s-1}{2}\right) &= \left\lceil \log_2\left(\frac{s-1}{2} + 1\right) \right\rceil \\
&= \left\lceil \log_2\left(\frac{s+1}{2}\right) \right\rceil \\
&= \lceil \log_2(s+1) - 1 \rceil \\
&= \lceil \log_2(s+1) \rceil - 1 \\
&= \text{bitcount}(s) - 1.
\end{aligned}$$

So bitcount(s) decreases by exactly 1. ✓
Again, subtracting 1 from an odd number clears the last bit and then dividing by 2 shifts right, so the bit count strictly decreases.

In both cases, bitcount(s) strictly decreases with every transition. Since bitcount$(s) \in \mathbb{N}$ (it is a nonneg integer), a strictly decreasing sequence of nonneg integers cannot continue forever. Therefore the algorithm always terminates. ■

(f) From part (e), bitcount(s) decreases by exactly 1 at every step. Starting from $(x, y, 0)$, the initial value of s is y, so the derived variable starts at bitcount(y) and decreases by 1 at each step until it reaches 0, at which point $s = 0$ and the algorithm halts. Therefore the algorithm terminates after exactly

$$\text{bitcount}(y) = \lceil \log_2(y+1) \rceil$$

steps. ■

Problem 4

Theorem. *If fewer than n students among those in an $n \times n$ arrangement are initially enlightened, and additional students reach enlightenment only by being taught by two or more neighbors, then there will be at least one student who never reaches enlightenment.*

Proof.

We use a monotonically non-decreasing derived variable argument. Consider the **perimeter** of the enlightened region, defined as the total number of edges shared between an enlightened cell and an unenlightened cell (or the boundary of the grid). More precisely, let

$$L(t) := \text{the number of edges on the boundary of the set of enlightened students on day } t.$$

Key Claim: $L(t)$ is non-increasing over time, i.e., $L(t+1) \leq L(t)$ for every day t.

Proof of Key Claim: Consider a single student who becomes newly enlightened on day $t+1$. This student must have had at least 2 enlightened neighbors on day t. We analyze how the perimeter changes when this student becomes enlightened. Suppose the newly enlightened student has $k \geq 2$ enlightened neighbors. Then:

- The k edges shared with enlightened neighbors are *removed* from the perimeter (they no longer separate an enlightened cell from an unenlightened one).
- The $4 - k$ edges shared with unenlightened neighbors (or the boundary) are *added* to the perimeter.

So the net change in perimeter from this one student is $(4 - k) - k = 4 - 2k$. Since $k \geq 2$, we have $4 - 2k \leq 0$. Therefore each newly enlightened student does not increase the perimeter, and so $L(t+1) \leq L(t)$. □

Bounding the initial perimeter: Each enlightened student is a unit square, contributing at most 4 edges to the perimeter. Therefore, if there are fewer than n initially enlightened students:

$$L(0) \leq 4 \cdot (\text{number of initially enlightened students}) < 4n.$$

Bounding the final perimeter: If *all* n^2 students were enlightened, the enlightened region would be the entire $n \times n$ grid, whose perimeter is exactly $4n$:

$$L(\text{all enlightened}) = 4n.$$

Conclusion: Since $L(t)$ is non-increasing and $L(0) < 4n$, we have $L(t) < 4n$ for all t. But if every student were enlightened, the perimeter would equal $4n$. Since the perimeter can never reach $4n$, the set of enlightened students can never be the entire $n \times n$ grid. Therefore, at least one student never reaches enlightenment. ■